

Nucleic Acids

Lecturer 7

Presented By

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Chapter V

RNA

Chapter Three: From gene to protein

- **1- The cell cycle & DNA Replication**

- **2- Gene Organization & Transcription**

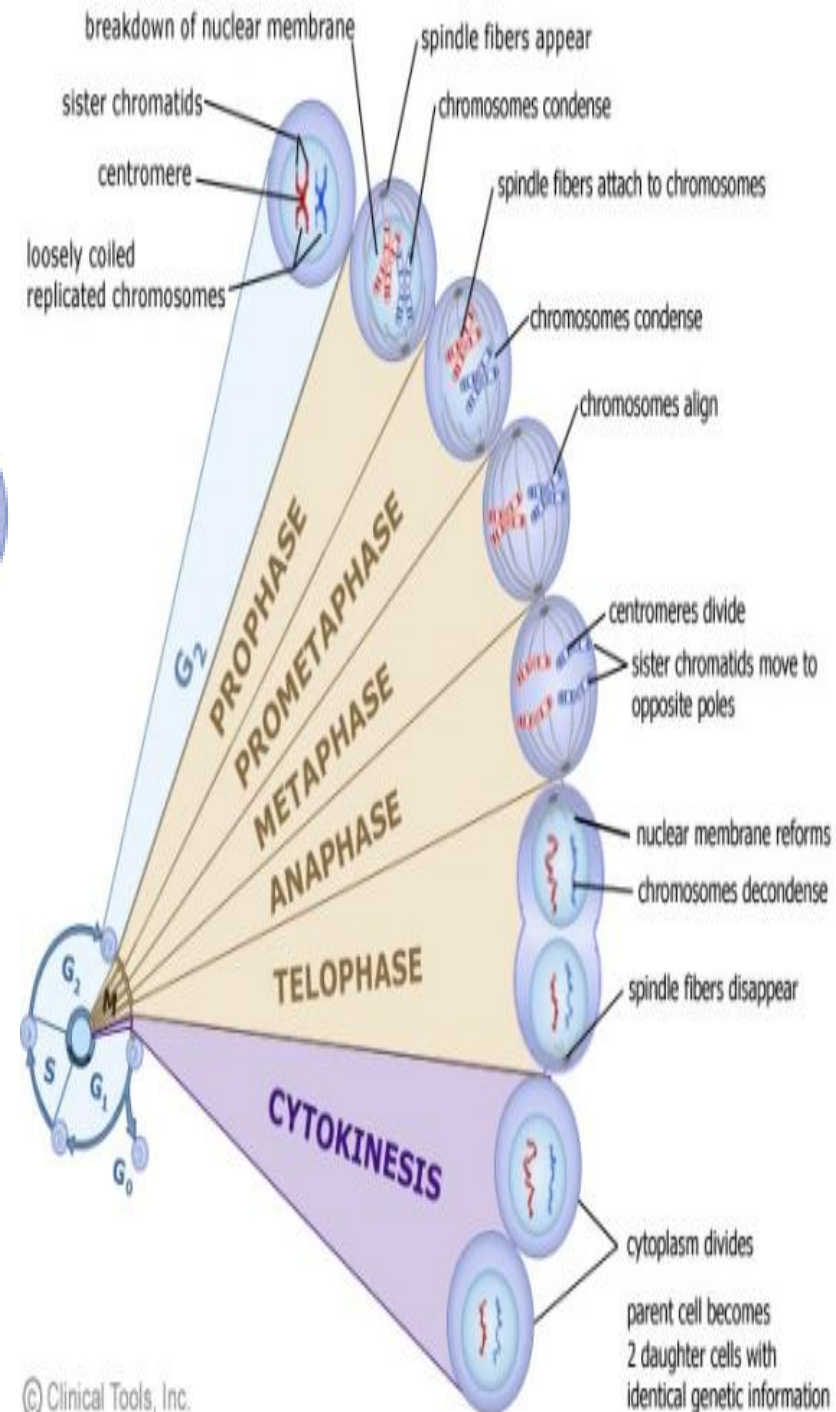
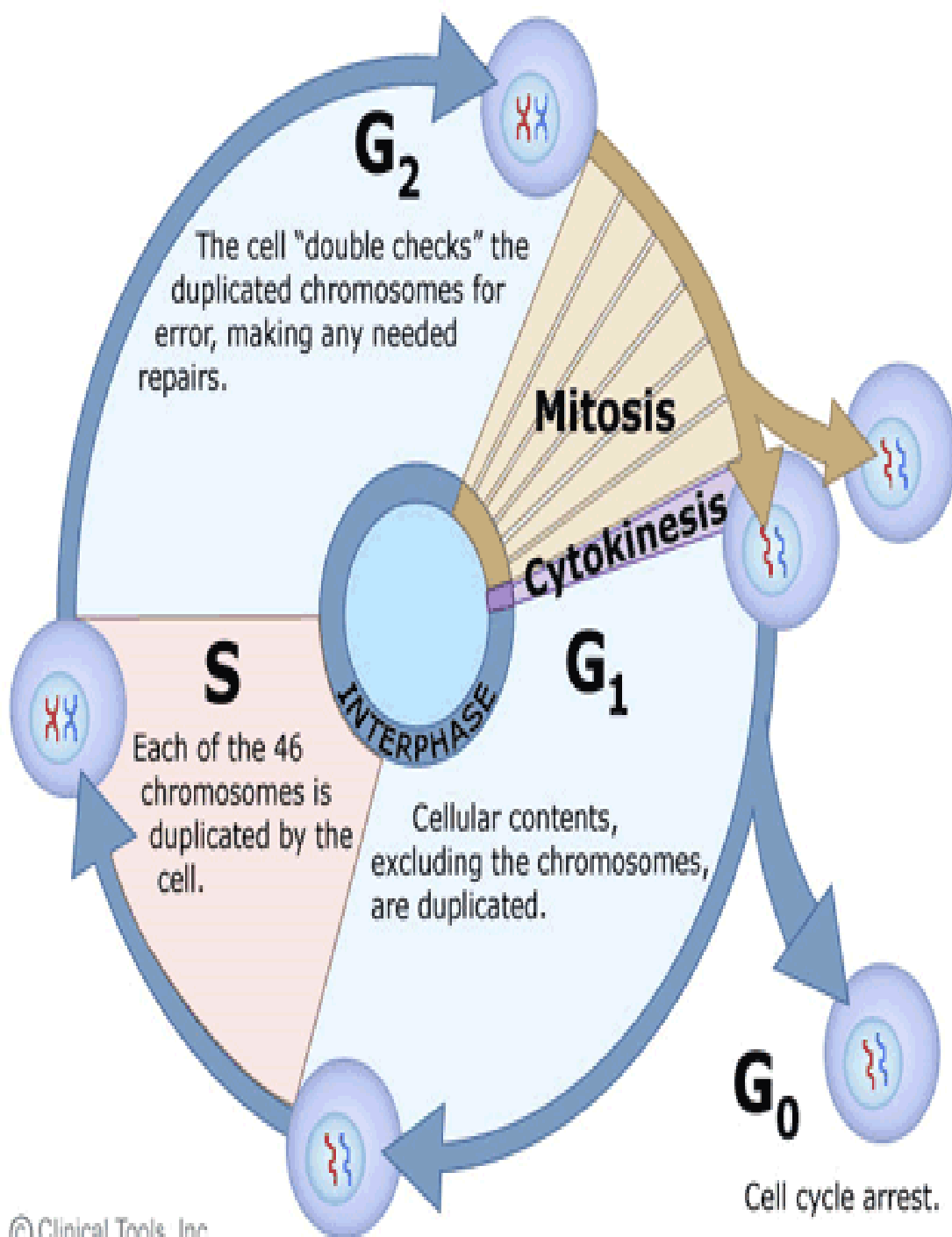
- **3- Genetic code & Protein Translation**

a- Cell Cycle:

- is the series of events that take place in a cell leading to its division and duplication of its DNA (**DNA replication**) to produce two daughter cells.
- Non-dividing cells not considered to be in the cell cycle.

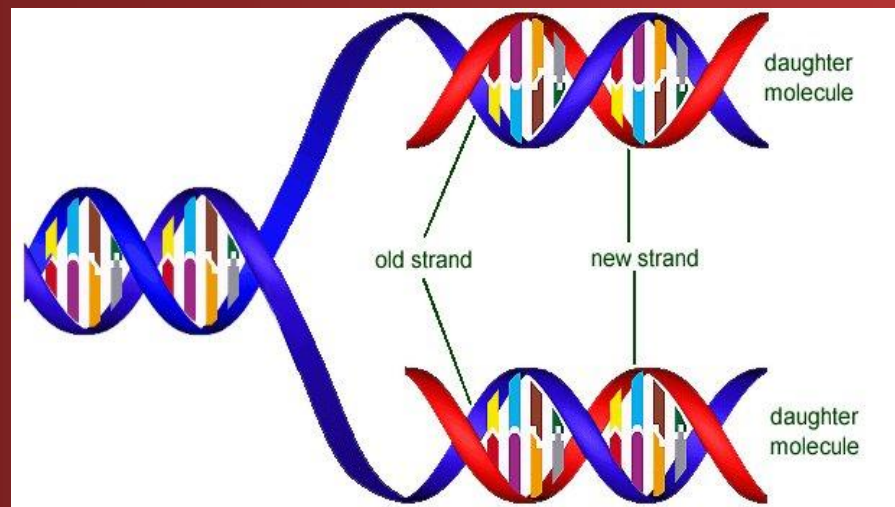
a- cell cycle stages:

- G1 stage stands for GAP 1: prior to DNA synthesis; 10 hrs.
- S stage stands for Synthesis: This is the stage when DNA replication occurs 9hrs.
- G2 stage stands for GAP 2: between DNA synthesis and mitosis); 4 hrs.
- M stage stands for mitosis: is when nuclear (chromosomes separate) and cytoplasmic division occurs. Mitosis is further divided into other phases.
- **G0:** cells which have stopped dividing



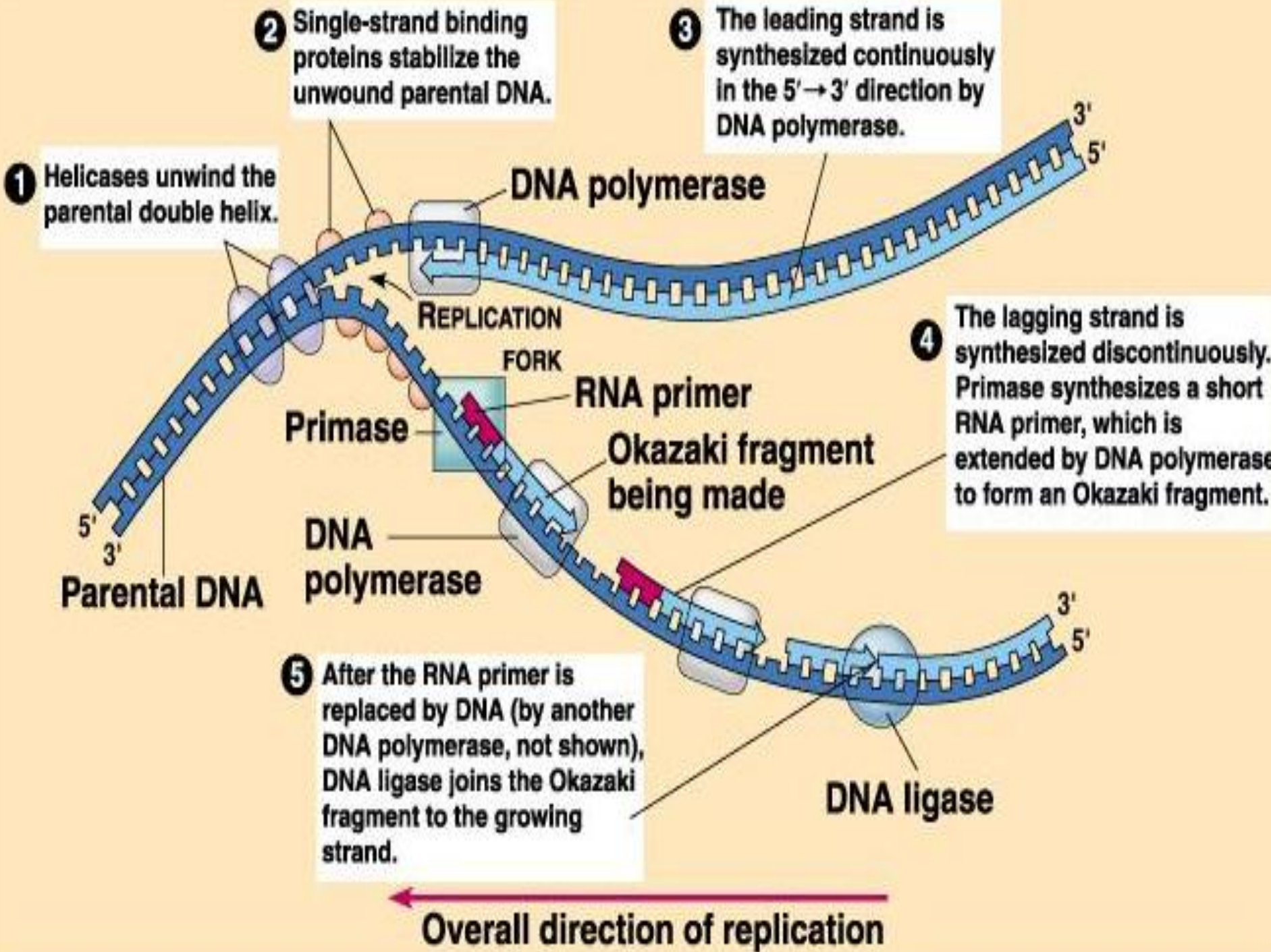
b- DNA Replication:

- Before each cell division, a faithful copy of the DNA needs to be made to ensure genetic continuity in the daughter cell.
- DNA replication is semi-conservative. Each daughter cell inherits one old and one new strand.
- The two strands are complementary to each other so each strand serves as a template for the synthesis of the other strand. This generates two identical copies.



b- DNA Replication:

- The DNA helix is very stable and has to be unwound before replication can occur. This is done by a DNA helicase, an enzyme that uses ATP as source of energy to break hydrogen bonds between base pairs.
- New DNA is synthesised by enzymes called DNA polymerases. DNA polymerases add nucleotides to the 3' end of a growing chain.
- DNA polymerases require a template strand, a primer (sequence of polymerized nucleotides made by a different enzyme), and a supply of deoxynucleotide triphosphates (dNTPs)

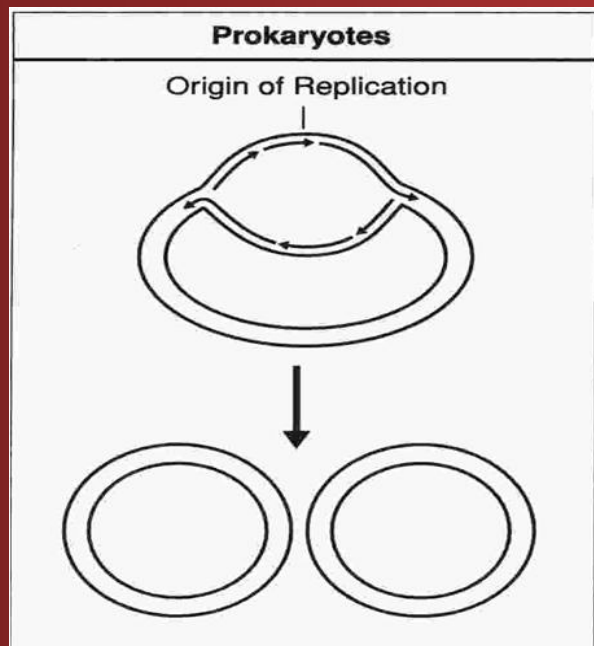


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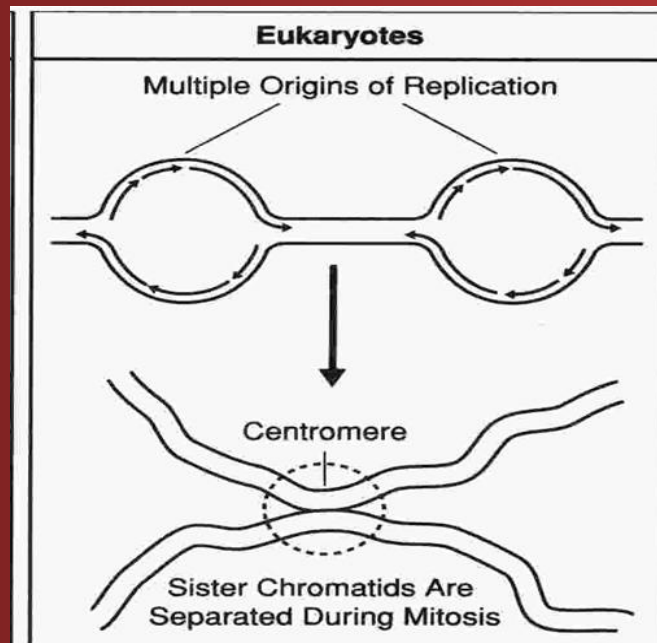
Replication of prokaryotic cells:

- Replication starts at a unique origin.
- The Two replication forks proceed simultaneously in opposite directions (creates a loop shapes of new DNA forming).
- The two forks meet at the other side of the circular chromosome. This is known as bi directional replication.



Replication of eukaryotic cells:

- Eukaryotic chromosomes are linear and very long.
- Multiple replication origins are distributed at intervals of about 100 kilobase pairs. Each replication origin gives bi-directional replication forks.
- Replication is finished when all the forks have met.



Eukaryotic chromosomes have multiple points of origin

